



basic education

Department:  
Basic Education  
REPUBLIC OF SOUTH AFRICA

**GEC PILOT STUDY  
MARKING GUIDELINE 2023  
NATURAL SCIENCES  
GRADE 9**

**SECTION A**

- One mark per answer.
- There are no half marks.

| No. | Expected answer (□) |   |
|-----|---------------------|---|
| 1.  | B                   | ✓ |
| 2.  | B                   | ✓ |
| 3.  | D                   | ✓ |
| 4.  | C                   | ✓ |
| 5.  | C                   | ✓ |
| 6.  | B                   | ✓ |
| 7.  | D                   | ✓ |
| 8.  | A                   | ✓ |
| 9.  | D                   | ✓ |
| 10. | B                   | ✓ |
| 11. | C                   | ✓ |
| 12. | B                   | ✓ |
| 13. | D                   | ✓ |
| 14. | B                   | ✓ |
| 15. | A                   | ✓ |
| 16. | C                   | ✓ |
| 17. | A                   | ✓ |
| 18. | B                   | ✓ |
| 19. | C                   | ✓ |
| 20. | D                   | ✓ |
| 21. | C                   | ✓ |
| 22. | A                   | ✓ |
| 23. | B                   | ✓ |
| 24. | A                   | ✓ |
| 25. | B                   | ✓ |
| 26. | C                   | ✓ |
| 27. | A                   | ✓ |
| 28. | C                   | ✓ |
| 29. | D                   | ✓ |
| 30. | B                   | ✓ |
| 31. | D                   | ✓ |

| No. | Expected answer (□) |   |
|-----|---------------------|---|
| 32. | C                   | ✓ |
| 33. | C                   | ✓ |
| 34. | B                   | ✓ |
| 35. | A                   | ✓ |
| 36. | D                   | ✓ |
| 37. | D                   | ✓ |
| 38. | A                   | ✓ |
| 39. | A                   | ✓ |
| 40. | D                   | ✓ |
| 41. | A                   | ✓ |
| 42. | A                   | ✓ |
| 43. | B                   | ✓ |
| 44. | B                   | ✓ |
| 45. | D                   | ✓ |
| 46. | C                   | ✓ |
| 47. | A                   | ✓ |
| 48. | B                   | ✓ |
| 49. | A                   | ✓ |
| 50. | A                   | ✓ |
| 51. | D                   | ✓ |
| 52. | D                   | ✓ |
| 53. | B                   | ✓ |
| 54. | C                   | ✓ |
| 55. | C                   | ✓ |

## SECTION B

- Do not penalize the learner for the same mistake more than once.
- There are no half marks.

| No. | Expected answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Clarification                                                                                                                                                                                                                                                                                                                                                                    | Mark |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1.  | <p>The <b>thicker</b> the wood (<b>greater</b> the distance)✓ between the magnet and the paper clips, the <b>weaker</b> the attraction force.✓</p> <p><b>OR</b></p> <p>The <b>thinner</b> the wood (<b>smaller</b> the distance)✓ between the magnet and the paper clips, the <b>stronger</b> the attraction force✓.</p> <p><b>OR</b></p> <p>The <b>closer</b> the magnet and the paper clips are to each other✓, the <b>stronger</b> the attraction force✓.</p> <p><b>OR</b></p> <p>The <b>further</b> the magnet and the paper clips are from each other✓, the <b>weaker</b> the attraction force✓.</p> <p><b>OR</b></p> <p>The <b>smaller/lesser</b> the distance between the magnet and the paper clips are from each other✓, the <b>stronger</b> the attraction force✓</p> <p><b>OR</b></p> <p>The <b>bigger/greater/larger</b> the distance between the magnet and the paper clips are from each other✓, the <b>weaker</b> the attraction force✓</p> | <p><b>Accept any correct and any other relevant responses.</b></p> <p><b>2 marks:</b> if any one fact is stated.</p> <p><b>0 mark:</b> if only part of the fact is provided and there is <b>no relationship between the thickness of the wood and the paper clips.</b></p> <p><b>NB: No mark for observation instead of conclusion. Observation is provided in the stem.</b></p> | 2    |
| 2.  | <p>2.1 Copper✓</p> <p>2.2 Nichrome✓</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p><b>1 mark:</b> for each correct answer</p> <p><b>0 mark:</b> incorrect answer</p>                                                                                                                                                                                                                                                                                             | 2    |

|     |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |   |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 3.1 | Parallel connection.✓                                                                                                                                                                                                                                           | <b>1 mark</b> for the correct answer                                                                                                                                                                   | 1 |
| 3.2 | As the number of cells increases, the potential difference remains the same.✓<br><br><b>OR</b><br><br>As the number of cells increases, the current remains the same.✓                                                                                          | <b>1 mark</b> for the correct reason<br><br><b>0 mark</b> for the correct reason with incorrect or no answer to the main question.                                                                     | 1 |
| 4.  | Cost = power consumption in kW x time in hours x unit price✓<br>= 1 kW x 2 h x R2/kWh✓<br>= R4✓<br><br><b>OR</b><br><br>Cost = power consumption in kW x time in hours x unit price✓<br>= 1 000 x 2 x 2✓<br>= $\frac{4\,000}{1\,000}$ (1 000 W = 1 kW)<br>= R4✓ | <b>3 marks:</b> correct calculation with correct formula and unit<br><br><b>2 marks:</b> correct calculations with no formula or correct calculation<br><br><b>1 Mark:</b> Correct formula/answer only | 3 |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 5.          | <p><b>Water (hydrosphere)</b> evaporates✓ and the <b>water vapour rises</b> to the atmosphere to form clouds✓</p> <p><b>OR</b></p> <p><b>Water vapour (atmosphere)</b> condenses to form clouds.✓ When the droplets (hydrosphere) become too heavy to be suspended they <b>fall as rain (hydrosphere)</b>✓</p> <p><b>OR</b></p> <p>Clouds produce rain that flows to rivers, streams and oceans</p> <p><b>OR</b></p> <p>Oceans absorb CO<sub>2</sub> from the atmosphere.</p> <p><b>OR</b></p> <p>Wave action adds O<sub>2</sub> to the water.</p> <p><b>Any other relevant correct explanation.</b></p> | <p><b>2 mark</b> for the correct explanation which has evidence of interaction between correct spheres as questioned.</p> <p><b>2 marks:</b> any correct or relevant explanation</p> <p><b>‘1 mark’ will NOT be awarded for any explanation</b></p> <p><b>0 mark:</b> If the learner shows no relationship in the response, there will be no mark allocated</p> <p><b>The event or activity mentioned should only reflect the two spheres.</b></p> | 2 |
| 6.          | <p>6.1 Sedimentary rock✓</p> <p>6.2 Layers of sediments✓</p> <p><b>OR</b></p> <p>Sediments</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p><b>1 mark:</b> each correct answer</p> <p><b>0 mark:</b> Layers of rocks<br/><b>Layers</b></p>                                                                                                                                                                                                                                                                                                                                                  | 2 |
| 7.          | <p><b>Acidic water</b> is formed✓ - <b>living organisms (natural vegetation and animals) die</b>✓</p> <p><b>OR</b></p> <p><b>The acid</b>✓ formed <b>corrodes</b>✓ parts of the infrastructures such bridges.</p> <p><b>OR</b></p> <p>Depleted oxygen✓ aquatic organisms die✓</p>                                                                                                                                                                                                                                                                                                                        | <p><b>2 marks:</b> correct explanation</p> <p><b>1 mark:</b> incomplete explanation of the impact stated.</p> <p><b>0 mark:</b> Pollutes the environment</p>                                                                                                                                                                                                                                                                                       | 2 |
| <b>[15]</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |